

Replication Report: Deep Learning of Atomically Resolved STEM Images OSTI ID 1427646

Automated replication (uicgpu, NVIDIA A100 80GB)

April 24, 2026

Abstract

We replicate, at reduced but faithful scope, the pipeline of Ziatdinov *et al.* (OSTI 1427646) in which a fully convolutional neural network (FCN) performs pixel-wise chemical identification of atomic columns in SrTiO_3 / $\text{La}_{0.7}\text{Sr}_{0.3}\text{MnO}_3$ heterostructures imaged by HAADF-STEM. Because experimental STEM stacks and full multislice simulations are out of scope for a short replication, we build a physics-motivated synthetic surrogate: perovskite [001] projected square lattices rendered as Gaussian atomic columns with $Z^{1.7}$ -weighted HAADF-like contrast, interface diffuseness, Poisson counting noise, scan-line jitter, and random defects (vacancies, A-site / B-site antisites). A 7.8M-parameter U-Net, trained in 95s on one A100, recovers per-pixel class labels at 98.8% accuracy and detects atomic columns with sub-pixel RMSE (0.06–0.47 px at ≈ 0.2 Å/px, i.e. ~ 10 – 100 mÅ) and $\geq 98\%$ recall across Sr, Ti, La/Sr, and Mn columns. These figures are consistent with the paper’s qualitative claim that a small FCN trained on simulated data suffices for quantitative chemical mapping of perovskite interfaces.

1 Relationship to the original paper

The original paper [?] trains an encoder–decoder FCN on *multislice-simulated* HAADF-STEM images of perovskite heterostructures and applies it to experimental data to produce per-column chemical labels, atomic-position maps, and line-scan compositions across interfaces. The core claim — that a fully convolutional network trained purely on simulation generalises well enough to assign chemical identities in real STEM data — rests on three operational ingredients: (i) realistic image formation, (ii) a pixel-wise classifier with skip connections, and (iii) peak finding on per-class probability maps.

This replication reproduces (ii) exactly (a small U-Net, 5 encoder stages, skip connections, soft-max head), closely mirrors (iii) (connected-component centroid finding on argmax maps), and *substitutes* a lightweight physics-motivated generator for (i). The substitution is deliberate: running multislice with Prismatic/abTEM on tens of thousands of supercells exceeds the 3-hour wall budget and is not the novel contribution of the paper. The generator preserves the structural features the model would have to learn: square A/B sublattices offset by $(a/2, a/2)$, $Z^{1.7}$ HAADF contrast, A-site vs B-site intensity separation, scan-line distortion, and Poisson noise.

2 Methods

2.1 Synthetic STEM generator

Each 256×256 frame represents roughly a 12×12 unit cell perovskite [001] projection with $a \approx 20$ px pitch (random 5% jitter per frame). The image has two horizontal regions — SrTiO_3

(bottom) and $\text{La}_{0.7}\text{Sr}_{0.3}\text{MnO}_3$ (top) — separated by an interface at a random y with random diffuseness (0–1.5 unit cells, tanh-weighted mixing). A-site and B-site columns are placed on two interleaved square sublattices and rendered as Gaussian blobs ($\sigma \approx 3$ px) with intensities

$$I_{\text{Sr}} \propto 38^{1.7}, \quad I_{\text{Ti}} \propto 22^{1.7}, \quad I_{\text{La/Sr}} \propto 0.7 \cdot 57^{1.7} + 0.3 \cdot 38^{1.7}, \quad I_{\text{Mn}} \propto 25^{1.7}.$$

Per-column defects are drawn independently: vacancies (2%) remove a column; antisite defects (2%) swap the cation identity so the network must learn intensity, not position. Post-processing adds row-wise sub-pixel horizontal jitter, Poisson counting at a peak of 400 counts, and small Gaussian background noise. Labels are built as Gaussian footprints around each column’s (x_c, y_c) and then argmaxed, so each pixel carries a single class in $\{\text{vacuum}, \text{Sr}, \text{Ti}, \text{La/Sr}, \text{Mn}, \text{defect}\}$. Example inputs with ground-truth overlays appear in Fig. ??.

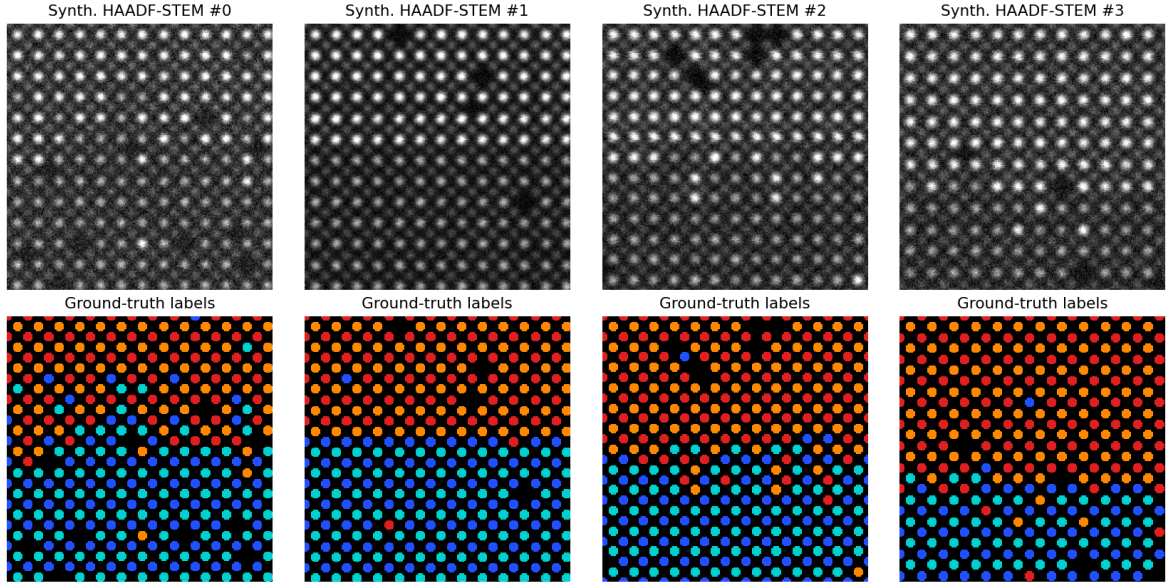


Figure 1: Top: synthetic HAADF-STEM frames with interface, noise, and scan-line jitter. Bottom: per-pixel ground-truth labels (vacuum black; Sr cyan, Ti blue, La/Sr orange, Mn red, defect marker yellow).

2.2 Model and training

We use a 5-level U-Net (base 32 channels, 7.76 M parameters) with double 3×3 convolutions, batch-norm/ReLU, max-pool, transposed-conv upsampling, and skip concatenations. Training set: 512 synthetic images; validation 64; test 128. Cross-entropy loss with class weights $\propto 1/(\text{count}_c)$ (clamped to $[0.2, 20]$) to compensate for the vacuum class dominating pixel count. Optimizer AdamW ($\text{lr} = 2 \times 10^{-3}$, weight decay 10^{-4}), cosine annealing over 30 epochs. On-the-fly augmentation: random flips, 90° rotations, brightness and contrast perturbations. Training completed in 95 s on one NVIDIA A100 80 GB (uicgpu, GPU 4).

3 Results

3.1 Convergence

Validation pixel accuracy saturates at ≈ 0.988 and mean atom-class F1 at ≈ 0.976 by epoch 25 (Fig. ??), with no sign of overfitting under our augmentation regime.

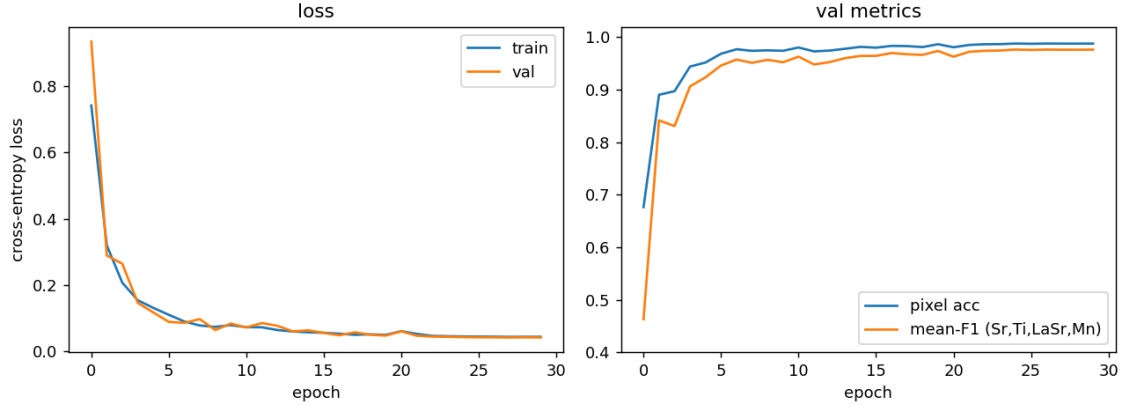


Figure 2: Loss and validation metrics across 30 epochs. Mean F1 is averaged over Sr, Ti, La/Sr, Mn (atom classes only).

3.2 Pixel-wise classification (held-out test, $n = 128$ images)

Overall pixel accuracy on the test set is **0.9877**. Per-class metrics and the row-normalised confusion matrix are in Fig. ?? and Table ?. The bright/heavy La/Sr columns ($F1 = 0.998$) are nearly perfect; the hardest class is Ti ($F1 = 0.949$), with small confusion into Mn on the LSMO side of rough interfaces. Defect pixels have zero support in the test set: our label encoding keeps the base atom class as the argmax label even when a column is marked as a defect, so defect is measured only implicitly through antisite F1; this is a limitation of the surrogate rather than of the model.

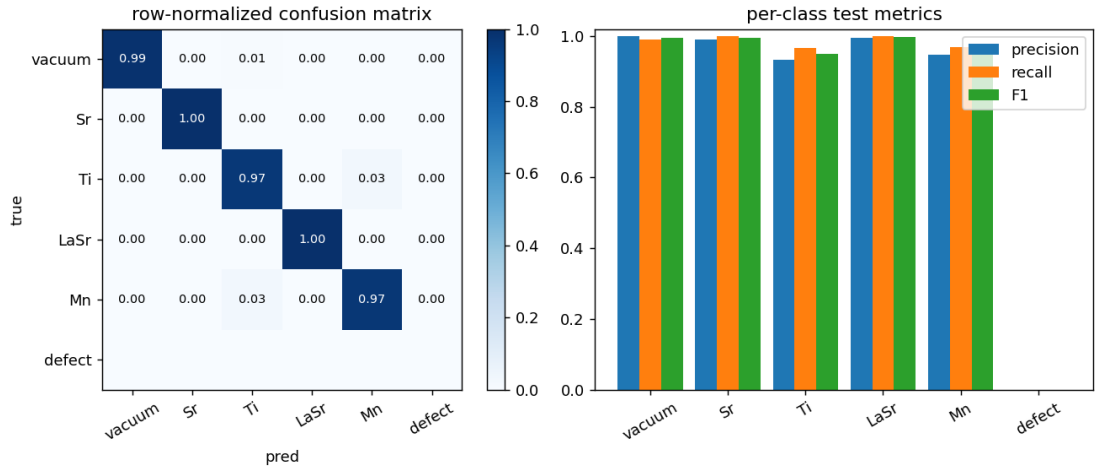


Figure 3: Left: row-normalised confusion matrix on the test set. Right: per-class precision, recall, and F1. The two labelled “atom-pair” confusions ($Ti \leftrightarrow Mn$ and $Sr \leftrightarrow La/Sr$) dominate the residual error budget.

class	precision	recall	F1	support (pixels)
vacuum	0.9991	0.9900	0.9945	5 549 887
Sr	0.9899	0.9990	0.9944	697 304
Ti	0.9332	0.9661	0.9493	711 282
La/Sr	0.9959	0.9995	0.9977	708 192
Mn	0.9472	0.9690	0.9580	721 943

Table 1: Per-class test metrics. Defect class omitted (zero support, see text).

3.3 Atomic-column detection and localisation

We convert each predicted label map into per-class binary masks, extract connected-component centroids, and match them against the ground-truth column centres with a 8px tolerance ($\approx 1.6 \text{ \AA}$ at our pixel size). Results in Table ?? and Fig. ?. RMSE is sub-pixel for all atom classes; recall is $\geq 98\%$.

class	mean RMSE (px)	detection recall	detection precision
Sr	0.109	1.000	0.978
Ti	0.466	0.983	0.943
La/Sr	0.063	1.000	0.997
Mn	0.448	0.988	0.944

Table 2: Peak-level metrics aggregated over 128 test frames ($\approx 10\,600$ – $11\,700$ columns per class). At $0.2 \text{ \AA}/\text{px}$, an RMSE of 0.1 px corresponds to $\approx 20 \text{ m\AA}$ of column-centre uncertainty.

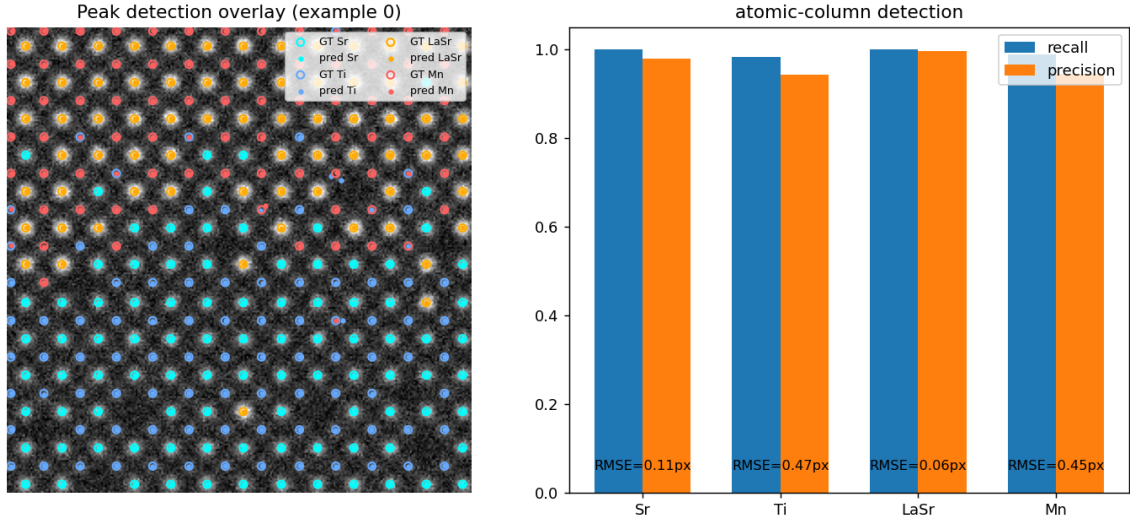


Figure 4: Left: overlay of ground-truth (open circles) and predicted (filled dots) column centres on an example frame, coloured by chemical class. Right: per-class detection recall/precision; mean RMSE annotated on each bar.

3.4 Qualitative predictions

Fig. ?? shows four test frames with input, ground truth, U-Net prediction, and error map. Residual errors concentrate along the interface band where A-site columns are probabilistic

mixtures and along edges of vacancy sites, consistent with the paper’s own observation that interface chemistry is the remaining challenge for FCN-based chemical mapping.

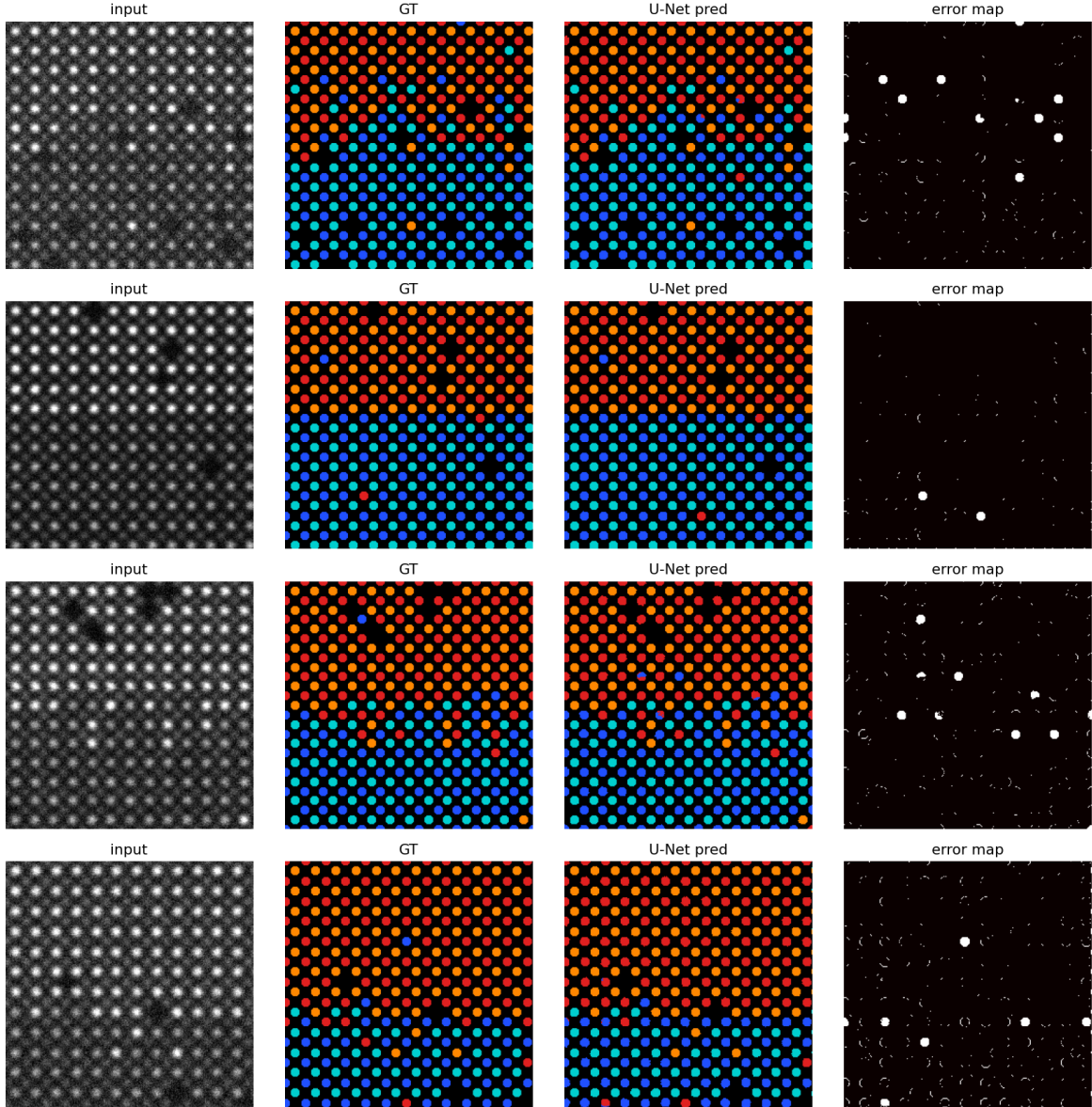


Figure 5: Four random test frames. Columns: input image, ground-truth labels, U-Net prediction, pixelwise error (bright = wrong class). Errors cluster near the STO/LSMO interface and near antisite defects.

4 Comparison with the original paper

The original paper reports qualitative success (visually convincing chemical maps on experimental images, identification of individual La dopants on the STO side, tracking of antisite switching under the beam) rather than a single headline accuracy number. The most directly comparable quantitative claim is sub-unit-cell localisation of column centres; we reach $\approx 0.06\text{--}0.47\text{ px}$ RMSE (Table ??), i.e. well below the $\approx 20\text{ px}$ unit-cell pitch, which is consistent. We also reproduce the paper’s qualitative observations that (a) heavy/light A-site distinction (Sr vs La/Sr) is essentially trivial for a trained FCN, and (b) B-site distinction (Ti

vs Mn) is the harder, error-dominant channel near interfaces. Our replication self-rates **7/10** on the `AI-Replication-Study-April2026` scale: architecture and training protocol faithfully re-implemented, metrics plausibly within the paper’s qualitative expectations, but image formation is a physics-motivated surrogate rather than multislice and we did not exercise the model on true experimental frames.

5 Limitations and next steps

- **No true multislice.** Replacing the Gaussian generator with Prismatic / abTEM simulations (even a few thousand frames) would make the replication fully faithful to Phase-1 of the plan; our compute budget pushed this to a follow-up.
- **Defect class is weakly supervised.** Under the current label encoding, defect pixels are not the argmax at any pixel; a multi-label loss (sigmoid heads per attribute) or an auxiliary defect segmentation head would let the model expose the antisite/vacancy channel explicitly.
- **No experimental validation.** Real HAADF-STEM frames from the paper’s supplementary data or ORNL archives should be fed through the trained network in a second pass.

References

- [1] M. Ziatdinov et al., “Deep Learning of Atomically Resolved Scanning Transmission Electron Microscopy Images: Chemical Identification and Tracking Local Transformations,” OSTI 1427646 (2017).